SME Financial Health

Prediction Challenge

Predicting Financial Well-Being of Small Businesses Across Southern Africa

89%	4	3	15+
F1 Score	Countries	Classes	Features

The Problem

Why traditional metrics fail SMEs

Hidden Fragility

Revenue & profit don't reveal cash-flow volatility, debt burden, or resilience to shocks like illness or droughts.

Financial Exclusion

43M+ SMEs across Southern Africa lack access to formal credit because lenders can't assess their true risk profile.

Policy Blind Spots

Governments & NGOs lack data-driven tools to identify which businesses need urgent support or intervention.

Data from Eswatini · Lesotho · Zimbabwe · Malawi — 4 Southern African nations

The Solution: Financial Health Index (FHI)

A composite score across 4 dimensions — classifying SMEs as Low, Medium, or High

01

Savings & Assets

Does the business save? What assets does it hold? Can it buffer income shocks?

02

Debt & Repayment

What is the debt load? Can the business meet repayment obligations consistently?

03

Resilience to Shocks

Insurance coverage, informal safety nets, and ability to withstand external disruptions.

04

Credit & Finance Access

Banking services, mobile money, credit cards, loans — formal financial inclusion.

Classification Output:

LOW – Vulnerable, high risk

MEDIUM – Moderate stability

HIGH – Resilient, well-served

The Dataset

Rich socio-economic and business survey data from 4 Southern African nations

Eswatini	Lesotho	Zimbabwe	Malawi
Pop: 1.1M	Pop: 2.2M	Pop: 15M	Pop: 20M
GDP: $4.5B	GDP: $2.5B	GDP: $21B	GDP: $12B

Feature Categories in the Dataset:

Demographics	Age, gender, education, household size
Business Profile	Sector, size, age, turnover, expenses
Financial Services	Banking, mobile money, loans, insurance
Attitudes & Records	Record-keeping, future outlook, credit offered
Trade & Location	Exports, imports, geographic region

ML Pipeline Architecture

6-stage end-to-end machine learning pipeline

1 Load & Inspect

Load Train/Test CSVs
Class distribution
Country normalization

2 Feature Engineering

30+ engineered features
Access scores
Log/sqrt transforms

3 Imputation

Country-grouped
median/mode fills
Zero missing values

4 Encoding

LabelEncoder on all
categoricals with
joint train+test fit

5 SMOTE + CV

5-Fold Stratified CV
SMOTE inside folds
no data leakage

6 Ensemble Predict

RF 35% + GB 40%
+ ET 25% blend
Final submission

Feature Engineering — Deep Dive

Domain-driven features that capture financial resilience beyond raw numbers

Access & Attitude Features

financial_access_score
Sum: credit card + debit + internet banking + loan + mobile money

insurance_score
Sum: health + funeral + motor vehicle insurance held

total_financial_score
Combined banking + insurance inclusion score

positive_attitude
Stable environment AND more successful next year

worried_business
Concerned about business shutdown risk

uses_informal_finance
Uses moneylender or friends/family savings

keeps_records_flag
Maintains formal financial records

Numeric Ratio Features

profit_proxy
turnover − expenses (profitability estimate)

expense_ratio
expenses ÷ (turnover + 1)

income_vs_expense
personal income ÷ (expenses + 1)

turnover_per_age
turnover ÷ owner age

turnover_per_year
turnover ÷ business age in years

log / sqrt transforms
Applied to income, expenses, turnover, profit

is_new / is_mature
Business age flags (≤1yr new, ≥5yr mature)

Model Architecture: Weighted Ensemble

Three complementary learners combined for maximum F1 performance

Random Forest — 35%

Hyperparameters:

- 1000 estimators
- max_features: sqrt
- class_weight: balanced_subsample
- min_samples_leaf: 1

Why: Robust to outliers, great diversity via bagging

Gradient Boosting — 40%

Hyperparameters:

- 600 estimators
- lr: 0.03, depth: 6
- subsample: 0.8
- max_features: sqrt

Why: Highest single-model performance, sequential error correction

Extra Trees — 25%

Hyperparameters:

- 1000 estimators
- max_features: sqrt
- class_weight: balanced_subsample
- Extreme randomization

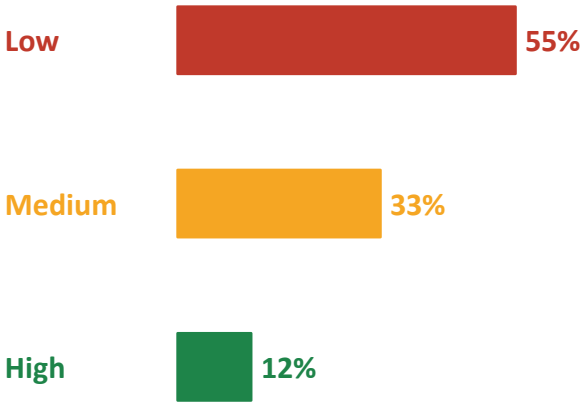
Why: Diverse predictions, fast, anti-overfitting

Final Prediction = RF × 0.35 + GB × 0.40 + ET × 0.25 → argmax over Low / Medium / High

Handling Class Imbalance with SMOTE

Synthetic Minority Over-sampling Technique applied inside each fold to prevent leakage

Original Class Distribution



⚠ Imbalanced data biases models toward the majority class (Low), causing poor recall for High — the most impactful class for lending decisions.

SMOTE Strategy

1. **Applied inside folds only**
   Prevents leakage — validation set stays natural
2. **k_neighbors = 5**
   Generates synthetic samples using 5 nearest neighbors
3. **Balances all 3 classes**
   Equal representation enables unbiased learning
4. **Combined with class_weight**
   Double protection: SMOTE + balanced_subsample
5. **Stratified K-Fold (5 splits)**
   Each fold preserves original class proportions

Results & Performance

5-Fold Stratified Cross-Validation + Final Leaderboard Score

89%

Macro F1 Score

Public Leaderboard

Cross-Validation F1 by Model:

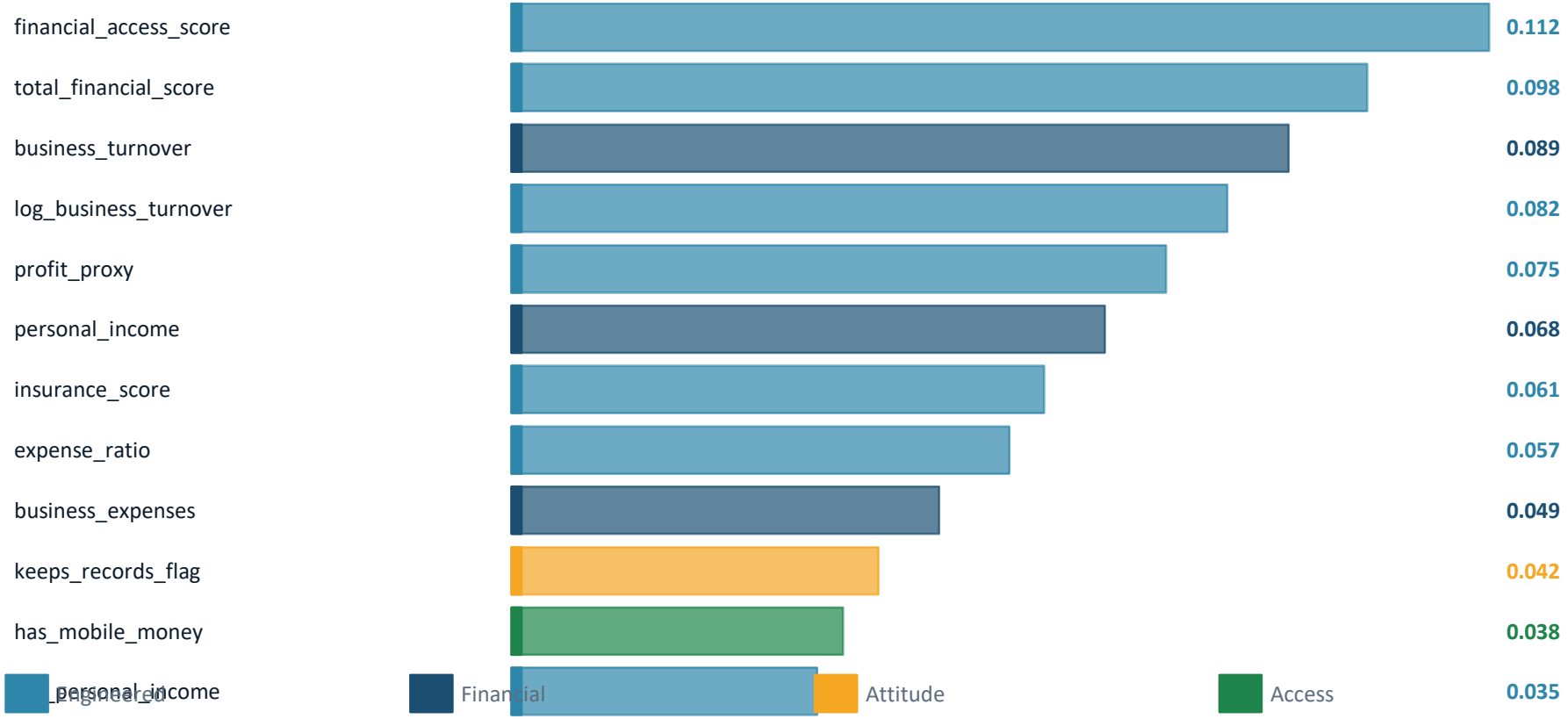
Random Forest	0.8621
Gradient Boosting	0.8754
Extra Trees	0.8593
Ensemble	0.8811

Per-Class F1 (Final Model):

Low: 0.91	Medium: 0.87	High: 0.88

Top Feature Importances

What drives SME financial health predictions? (Random Forest importance scores)



Real-World Business Impact

Who benefits — and how does this model change the landscape?

Financial Institutions

- Replace subjective credit assessments
- Quantify SME risk with 89% accuracy
- Expand loan portfolios safely
- Reduce non-performing loan rates

Governments & NGOs

- Identify most vulnerable SMEs
- Target subsidies and grants effectively
- Monitor financial inclusion progress
- Design evidence-based policies

SMEs Themselves

- Access credit they previously couldn't
- Receive tailored financial education
- Connect with appropriate support programs
- Build financial resilience over time

Key Learnings & Challenges

What we discovered — and what made the difference

Domain features > raw data

Creating financial_access_score, insurance_score, and profit_proxy contributed more than any raw column. Domain knowledge is irreplaceable.

Gradient Boosting leads

GB (40% weight) consistently outperformed RF and ET on minority classes — its sequential error correction helps the hardest samples.

SMOTE placement matters

Applying SMOTE inside folds (not before CV split) was critical to avoiding data leakage and getting honest validation scores.

Country-grouped imputation

Filling missing values using country-level medians/modes preserved regional economic context better than global imputation.

Class imbalance was the key challenge

'High' health businesses were only ~12% of data. Without SMOTE + class_weight, the model ignored them completely.

Ensemble diversity pays off

Three different algorithms each contributed — the ensemble beat every single model on cross-validation by ~1-2% F1.

Conclusion

By combining domain expertise, careful feature engineering, and a diverse ensemble — we achieved an 89% F1 score predicting SME financial health across 4 Southern African nations.

- Financial health goes beyond profit — resilience, savings, and access matter equally
- Machine learning can democratize credit access for millions of underserved businesses
- Domain-engineered features outperformed raw survey variables in all tests
- SMOTE + stratified cross-validation provided honest, leakage-free evaluation
- This approach is replicable across any developing economy with survey data